

KAPIL

Dehradun, Uttarakhand, India

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[Portfolio](#)

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EDUCATION

University of Petroleum and Energy Studies

B.Tech in Computer Science and Engineering — CGPA: 7.6/10

2023 – 2027

Dehradun, Uttarakhand

EXPERIENCE

Weazy - Web Developer Intern

June 2025 – Aug 2025

Remote

- Engineered a full-stack Flask REST API automating product image fetching, processing, and database updates for an e-commerce catalog, reducing manual catalog effort by 60%
- Implemented intelligent web scraping pipelines with Selenium and BeautifulSoup to fetch relevant images from Google, Bing, and DuckDuckGo with filtering, preview, and selection logic.
- Architected a cloud-ready microservices solution with modular design, scalable file handling, and environment-based configuration; deployed on Azure with CI/CD pipeline support.

PROJECTS

RAG Based Assistant

LangChain, FastAPI, Docker, Kubernetes, FAISS

- Engineered a scalable, containerized RAG application using LangChain, FastAPI, and FAISS vector database with REST API endpoints, supporting 50+ concurrent users.
- Designed a multi-stage query pipeline prioritizing local knowledge base, with **LLM Fallback** to **Wikipedia API** and LLM generation, boosting answer accuracy by 35%.
- Orchestrated deployment with Docker Kubernetes, reducing setup time by 40% and improving maintainability..

NovelNet

Django, Supabase, HTML, Google Books API

- Designed and deployed a full-stack Django book review and recommendation platform serving 100+ users, integrating Google Books API, JWT authentication, and cloud file management.
- Implemented secure file handling with tracking, user attribution, and REST API endpoints for mobile integration using Django REST Framework
- Streamlined deployment and scalability using Gunicorn and PostgreSQL, ensuring secure authentication and cross-platform performance optimization

VesselVoice

ResNet, Signal Processing, Librosa, REST API

- Built a multi-class audio classification pipeline using CNN and ResNet-18 with deep learning-based feature extraction, achieving classification across 5 categories using spectrogram analysis
- Engineered a custom feature extraction pipeline (MFCCs, spectral features) and built a real-time inference engine with sliding window segmentation and probability averaging for robust predictions
- Architected a FastAPI backend with real-time predictions and a React dashboard, containerized with Docker.

AEGIS

PostgreSQL, Distributed Systems, API Gateway

- Implemented distributed consensus system for GPU inference orchestration handling 100+ requests per second with fault-tolerant task scheduling and high availability across multiple GPU nodes..
- Developed multi-backend inference routing with automatic failover across vLLM, llama.cpp, Ollama, and HuggingFace API using circuit breaker and load balancing patterns.
- Implemented enterprise security and monitoring with PostgreSQL audit logging, JWT authentication, Prometheus and Kubernetes.

TECHNICAL SKILLS

Languages: Python, C++

ML/AI: TensorFlow, PyTorch, scikit-learn, OpenCV, NLTK, LangChain, Prompt Engineering, RAG, Fine-tuning

Web Dev: Django, FastAPI, Pydantic

DevOps: Docker, Kubernetes, CI/CD (GitHub Actions, GitLab, Jenkins), Azure VM, Azure ML, Cloud Deployment

Databases: PostgreSQL, MongoDB, MySQL, FAISS, Vector Databases

ACCOMPLISHMENTS & CERTIFICATIONS

- Reached Final Round of Deloitte Hacksplasion, ranked among Top 60 teams nationwide.
- Cleared Coding Rounds I & II and advanced to interview stage of HackWithInfy.
- Oracle Data Platform 2025 Certified Foundations Associate - Oracle
- Introduction to Generative AI – Coursera